

## Team 10 — DineCheck (Campus Dining System): One-Page Budget

Revised October 5, 2026. Loaded rate: \$60/hour, the instructor-provided rate for CS 4390/5388.

### 1. Write the assumptions

| Cost Driver / Assumption                                                                                 | Owner           |
|----------------------------------------------------------------------------------------------------------|-----------------|
| Campus dining/menu data is available and does not require a complex outside integration.                 | Developer       |
| Web and mobile use one responsive application instead of separate native apps.                           | Developer       |
| Crowd levels and estimated wait times remain outside this project slice.                                 | Project manager |
| Testing and deployment can be completed using the team's existing development environment.               | Developer       |
| Project scope stays limited to dining locations, hours, menus, dietary filtering, and item availability. | Project manager |

### 2. Convert and staff

Planning estimate: 694 team hours (Session 9, PERT)  $\div$  130 = 5.34 FTE-months.

| Role            | Month 1     | Month 2     | Total       |
|-----------------|-------------|-------------|-------------|
| Project Manager | 0.17        | 0.03        | 0.20        |
| Developers      | 2.68        | 1.22        | 3.90        |
| Testing         | 0.25        | 0.42        | 0.67        |
| UX / Mobile     | 0.41        | 0.16        | 0.57        |
| <b>Total</b>    | <b>3.51</b> | <b>1.83</b> | <b>5.34</b> |

### 3. Price it

| Role                   | FTE-months          | Loaded Rate | Cost            |
|------------------------|---------------------|-------------|-----------------|
| <b>Project Manager</b> | 0.20 (26 h)         | \$60/hour   | \$1,560         |
| <b>Developers</b>      | 3.90 (507 h)        | \$60/hour   | \$30,420        |
| <b>UX / Mobile</b>     | 0.57 (74 h)         | \$60/hour   | \$4,440         |
| <b>Testing</b>         | 0.67 (87 h)         | \$60/hour   | \$5,220         |
| <b>Labor Total</b>     | <b>5.34 (694 h)</b> |             | <b>\$41,640</b> |

Non-labor: hosting \$150/month  $\times$  2 = \$300; tools \$300; domain/deployment \$100; database/cloud \$300; total \$1,000. Base build estimate = \$41,640 + \$1,000 = \$42,640.

Operating cost per year: hosting \$150  $\times$  12 = \$1,800 + maintenance 4 h/month  $\times$  12  $\times$  \$60 = \$2,880 = \$4,680.

### 4. Price the risks

| Risk                                                  | Chance | Extra hours | EMV   |
|-------------------------------------------------------|--------|-------------|-------|
| Menu data needs extra cleanup or access work          | 30%    | 24          | \$432 |
| Stale availability data needs more testing and rework | 25%    | 20          | \$300 |
| Requests grow the agreed slice                        | 20%    | 30          | \$360 |

EMV = chance  $\times$  extra hours  $\times$  \$60. Contingency reserve = \$1,092. Cost baseline = \$42,640 + \$1,092 = \$43,732. Management reserve (5%) = \$2,187. Total budget request = \$45,919.